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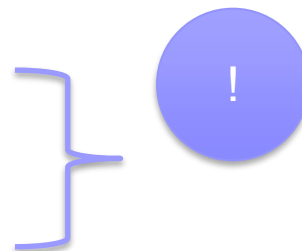
RDF & SPARQL: Exercises Exercises



dat·AI data * AI
AI * data

Overview

- Recap on RDF
- Recap on SPARQL



Indicates content that is more interesting from a pragmatic point of view and/or novel w.r.t. theory slides

Separate presentation

- Exercises in RDF + SPARQL
 - For RDF exercise this year also consider
 - DS2526 - ESE1.0 - Exercises about RDF modeling
- Exercises in Advanced SPARQL

Exercises

RDF Exercise 2025-2026

- This year, the main exercise on RDF modeling is in files:
 - DS2526 - ESE1.0 - Exercises about RDF modeling – EN
- RDF exercises in this presentation
 - In class
 - Left for additional homework / additional training / self assessment

Exercise no. 1 // RDF



- Exercise #1 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL”
 - In class: warm-up

Exercise no. 2 // RDF



- Describe in RDF the following statements:
 - Dipartimento di Informatica, Sistemistica e Comunicazione of Università degli studi di Milano – Bicocca is located in Viale Sarca 336, Milan
 - The Web site <http://disco.unimib.it> contains the activity of Dipartimento di Informatica, Sistemistica e Comunicazione
 - (a) Write these statements in RDF in N3, **Turtle** and RDF/XML format without blank nodes, using at least three triples for each statement.
 - (b) Write these statements in RDF in N3, **Turtle** and RDF/XML format with blank nodes, using at least three triples for each statement.
 - (c) Draw the graph of the exercise.

This can be replaced by exercise in [DS2526 - ESE1.0 - Exercises about RDF modeling](#)

I leave a solution here as a reminder to include the actual graph drawing as part of these exercises

Possible solution Turtle (b)

@prefix ex: <http://disco.unimib.it#> .

ex:disco ex:name "Dipartimento di Informatica, Sistemistica e Comunicazione" .

ex:disco ex:department ex:unimib .

ex:unimib ex:name "Università degli studi di Milano – Bicocca" .

ex:disco ex:address [ex:street "Sarca, 336";

ex:CAP "20126";

ex:municipality "Milano";

ex:country "Italia"] .

@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .

@prefix : <http://www.example.com#> .

<http://disco.unimib.it.org/> rdf:type :website .

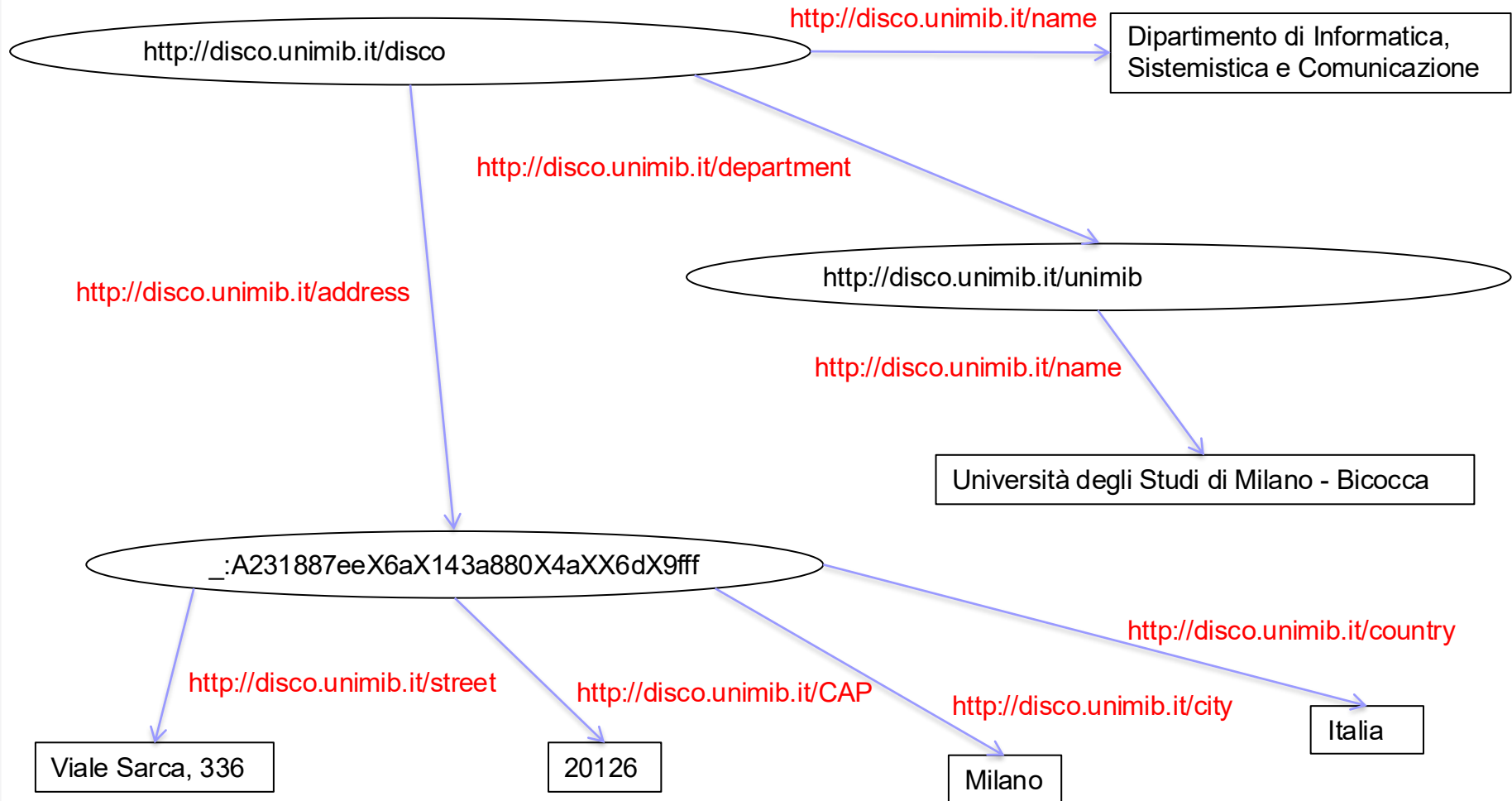
<http://www.disco.unimib.it> :hosts :disco-activity .

:disco-activity :has-homepage [rdf:type :homepage;

url "http://disco.unimib.it"] .

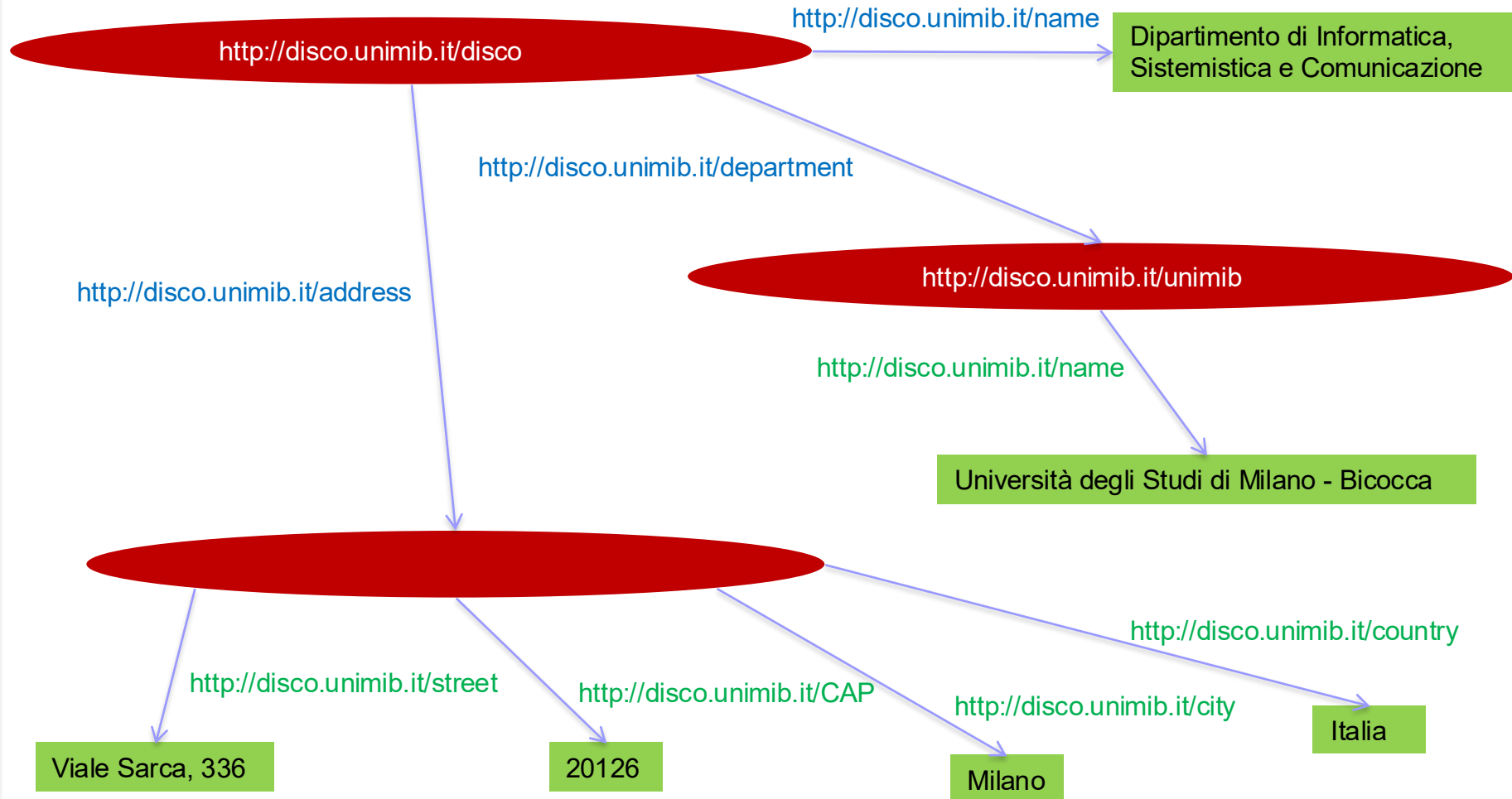
RDF Graph for (c)

Drawing the graph is very useful to learn to represent using a graph model



RDF Graph for (c)

Colors highlight literals (in green) and distinguish edges connecting IRI/BNs to IRI/BNs vs IRI/BNd to literals





Exercise no. 3 // RDF and Modeling

- Exercise #3 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL”
 - Some exercise with reification
 - useful for practicing with blank nodes and the triple structure
 - ~ treat a statement as an entity
 - E.g., Mat believes that Marco is a nice guy
 - Mat believes A . A = “Marco is a nice guy”
 - Do the best you can / think how to do it in the given time
 - Rest of the exercise at home



Exercise no. 4 // RDF and Modeling

- Exercise #2 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL”
 - ☐ As before: some exercise with reification
 - ☐ Do the best you can / think how to do it in the given time
 - ☐ Discussion
 - ☐ Rest of the exercise at home

Exercise no. 5 // SPARQL



- Exercise #4 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL”

Exercises no. 6 // SPARQL Details

- Given the query in natural language write the corresponding SPARQL query
- Use as endpoint
<https://dbpedia.org/sparql>



How many employees does Google
have?



How many employees does ~~Google~~
Rolex have?



Retrieve movies that were written,
directed or produced by Charlie
Kaufman

**NEXT QUERIES DEPENDING
ON THE TIME... LIKELY ●**

In which country does the Nile
start?



Give me a list of all trumpet players
that were bandleaders.

Give me the number of all trumpet players that were bandleaders.

Which countries have places with more than two caves?

Which software has been developed by organizations founded in California?

Was Natalie Portman born in the
United States?

Is Rome the capital of Italy?

Does Elizabeth Tylor have a
birthdate in DBpedia?

Is Milan a city in Germany?

Is Michelle Obama the spouse of
Barack Obama?

Build a graph such that all entities
heavier than 1000000000 are
American basketball players

Buld a graph that associates the
type **dbo:Aviator** to all those
entities that have **dbr:airline** as
occupation in DBpedia

Describe all entities that have
dbr:airline as occupation

Please describe all subjects and
objects of the **dbo:branchFrom**
relation

Exercise no. 7 // More SPARQL

- Exercises #5-#12 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL” contain many more queries with specific features and solutions.

References

- <https://www.w3.org/TR/rdf-sparql-query/>
- <https://www.w3.org/RDF/>
- <https://wiki.dbpedia.org/>

THANK YOU